Tracking Error as an Observation for Behavior Cloning

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Abstract: The ACT configuration studied here exposes one timestep of measured joint positions to the policy. We show that appending the servo tracking error, the difference between the last applied position command and the current measured position, changes what a policy can learn at a 50-demonstration budget. On a physical SO-101, after a fixed demonstration replay, three training seeds were trained and evaluated in paired trials. Tracking error wins each of the two completed 20-pair comparisons, 9 vs 5 (exact McNemar $p = 0.29$) and 14 vs 1 ($p = 0.001$); the third evaluation was interrupted by a gripper fault after six pairs. Pooled, the completed seeds place 23 of 40 against 6 of 40 (post hoc stratified exact test $p = 0.0005$). In simulation over five seeds, placement rises from 6.2% to 21.8%, and a position-history control recovers most of that gain, consistent with a benefit from temporal information without establishing a preferred representation. The signal requires no added sensor: it is reconstructible from logs with aligned commands and positions, and it keeps varying where the servo's load register saturates at a configured torque limit in 16.5% of demonstration frames. Across 546 archived teleoperation episodes, a preregistered prediction that larger tracking error at grasp onset accompanies a gripper-retention proxy fails (pooled $d = -0.39$). The study motivates reporting observation history, applied-command semantics, and actuator limits as experimental choices.

Keywords: Behavior Cloning, Proprioception, Low-Cost Manipulation

1 Introduction

The proprioceptive input to a behavior-cloning policy is a consequential design choice. Our ACT [1] implementation in LeRobot [2] reads current measured joint positions and no proprioceptive history. On a position-controlled hobby-servo arm, a jaw can stall against an object while its command continues closing. The gap between command and position then carries information about resistance, mixed with actuator dynamics. Commands appear in the action labels, but their difference from measured positions is absent from the policy input.

This paper treats that default as an object of study, in the way recent work has treated controller gains and action spaces [3, 4]. We append the tracking error, $\delta_t = g_{t-1} - q_t$, to the observation of an otherwise unchanged ACT policy and measure what changes. The feature needs no sensor, force model or calibration, and it can be reconstructed from any recording whose logged commands are the goals the servos received. It also mixes contact with following lag and actuator dynamics, and a short history of positions carries much of the same information. The questions are whether exposing it helps, whether the help is specific to the command channel, and what a paper should report so that the answer transfers.

We answer with three experiments. A simulation grid sets explicit tracking error against action history, position history, and a compensated residual (tracking error minus a fitted free-motion term) under identical data and training budgets. A hardware study trains ACT on 50 human demonstrations on one SO-101 at three training seeds and evaluates each seed in paired trials; two evaluations

reached their 20-pair budget and the third was interrupted by a gripper fault. A corpus study asks, on 546 archived episodes, whether the same signal can serve as an outcome label. Tracking error wins both completed hardware comparisons, 9 vs 5 and 14 vs 1 of 20. In simulation, position history recovers most of the mean gain, with substantial uncertainty in their difference. The contribution is a controlled observation-design study: tracking error improves this hardware comparison, while temporal controls limit a feature-specific interpretation.

2 Related work

Hwang et al. [5] estimate torque on RC servos by system identification, FACTR 2 [6] learns external-torque estimates from free-motion data and resamples training data by force, and NeuralActuator [7] fits an actuator model on the SO-101 from command, position, tracking error, and the load register and feeds its estimated force to a policy. We use the raw command–position difference with no model and make correspondingly weaker claims about force. Force-aware imitation on sensed arms finds that how the signal is presented matters [8, 9].

Wong et al. [10] show that leader–follower discrepancies in teleoperation carry interaction cues: removing them from ACT's action labels degrades force-critical phases, while torque proxies restore contact-aware behavior, and Xu et al. [11] condition policies on command–state history with latency-aware trajectory inpainting. We study a single-step discrepancy input to ACT with fixed command targets and execution settings, including temporal controls and a telemetry-saturation audit. Adding the feature also adds access to the previous command, and history that helps under partial observability can equally support a shortcut that copies the previous action [12, 13, 14]. We therefore compare several temporal observations rather than reading any gain as contact sensing.

3 Observation and experimental design

Let q_t be the measured joint position and g_{t-1} the preceding applied position goal, expressed in the same coordinate system. We define

$$\delta_t = g_{t-1} - q_t, \qquad o_t^{\text{base}} = [I_t, \widetilde{q}_t], \qquad o_t^{\delta} = [I_t, \widetilde{q}_t, \delta_t/c], \tag{1}$$

where I_t is the camera observation and $\widetilde{q}_t$ uses training-set normalization. Positions are degrees for the arm joints and a 0–100 opening scale for the gripper on hardware, integer encoder counts in simulation; $c = 8$ and the command limiter are in those units, and c is not a torque conversion. Under approximately static contact, tracking error reflects load on a position servo; during motion it also contains following error, and we use it as an observation without assigning every nonzero value to contact. Given q_t, tracking error and the previous command contain identical information: $g_{t-1} = q_t + \delta_t$. Their comparison therefore tests representation and scaling; position history tests whether commands are needed at all.

Simulation. We train ACT policies [1] on 280 scripted pick-and-place demonstrations with randomized object and scene properties, in a MuJoCo model of the arm with position actuators and encoder counts. The scripted demonstrator regulates its jaw from the same tracking error, which favours designs that observe it (Appendix E). All designs share the same data and training budget: 12,000 steps with 96-pixel front and gripper-camera images. Each policy is evaluated on 100 episodes with shared scene seeds across designs, with five training seeds for the original designs and three for the position-history control. The comparison uses placement success with the crush threshold disabled, so it tests grasping and transport rather than regulation within a force band.

Hardware. We collect 50 teleoperated demonstrations of placing an adapter inside a tape roll using one SO-101 and one overhead camera. Both policies use the same 18,549 training frames after removal of stationary prefixes, 96-pixel images, and 6,000 training steps. The proprioceptive input has six dimensions for the base and twelve with tracking error; every other architecture and optimization setting is shared. Three seeds are trained, one checkpoint per observation design and seed, and every checkpoint is evaluated.

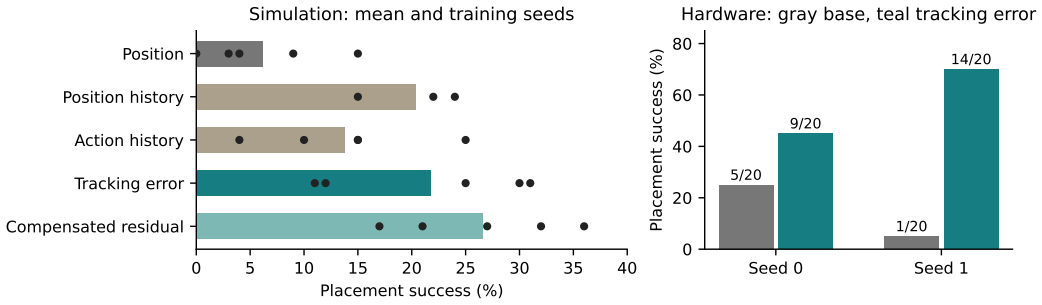


Figure 1: Placement success by observation design in simulation and in the two completed hardware evaluations. Left: bars are means; dots are individual training seeds (five per design, three for position history). Right: hardware success counts for the two completed evaluations. The full evaluation record, including an interrupted third comparison, appears in Appendix A.

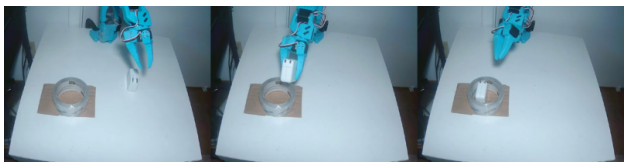


Figure 2: Three frames from a tracking-error policy (seed 1) after the scripted replay, from a showcase rollout that is not included in any count.

Evaluation begins after replaying 115 commands from a fixed demonstration, which places the arm near the grasp, after which each policy controls up to 400 steps. In the demonstrations the grasp closure occurs at a median frame 331 of a median 400-frame episode and after frame 115 in 48 of the 49 that close, and a median 67% of joint travel follows frame 115, so the replay covers the approach and leaves the grasp to the policy (Appendix B). ACT predicts and executes 30-command chunks, so observations inform the next chunk rather than each command within it. A relative-target limiter is shared by both policies, and the tracking-error input uses the command returned *after* limiting. Trials are paired by training seed and physical reset: the operator returns the adapter and roll to their nominal start positions by hand, base-first and tracking-error-first order alternates across pairs, and the operator, who is not blinded, records whether the adapter rests inside the roll (Appendix A).

4 Results

Hardware: tracking error wins each completed paired comparison. Three seeds were trained with the frozen recipe; seeds 0 and 1 completed their 20-pair budgets and seed 2 was interrupted. Table 1 reports each completed seed as its own paired comparison, which is the registered analysis. Tracking error improves placement by 20 percentage points for seed 0 and by 65 points for seed 1. Exact two-sided McNemar tests use only the discordant pairs: seed 0 has six pairs favouring tracking error and two favouring the base ($p = 0.29$), seed 1 fourteen and one ($p = 0.001$). Seed 0 alone does not meet the registered criterion; seed 1 does. As a post hoc summary, pooling the two completed seeds gives 23 of 40 placements against 6 of 40, with 20 discordant pairs favouring tracking error against 3 (stratified exact test $p = 0.0005$); the conditional odds ratio is 6.7 (95% CI 2.0 to 35.0), and the equal-weight mean is 42.5 points over the completed seeds. These tests describe trial-to-trial variation for fixed checkpoints, not variation over training seeds; with 20 pairs the exact test has power 0.17 under seed 0's discordance structure and 0.97 under seed 1's (Appendix A). A third evaluation was interrupted by recurring gripper faults after six all-failure pairs (Appendix A). Counted by policy over every hardware record (Table 4), the gripper overload at shutdown struck 4

Table 1: Hardware placements and paired outcomes for the two completed evaluations. Discordances list tracking-error-only / base-only successes. See Appendix A for the interrupted evaluation.

Seed	Evaluation	Base	Tracking error	Discordances	p
0	Complete	5/20	9/20	6 / 2	0.2891
1	Complete	1/20	14/20	14 / 1	0.0010

of 47 position-only, 6 of 55 tracking-error and 0 of 10 position-history rollouts, so it is a property of the bench rather than of one observation design.

The checkpoints read the channel: zeroing the tracking-error input offline moves the predicted jaw command by 12.6 to 13.0 units on frames where the gripper load register is pinned and by 2.7 to 2.9 units elsewhere, against a mean recorded command step of 0.5 (Appendix C). The dependence concentrates where the jaw is loaded, without showing that it produced the placements.

Simulation: temporal controls limit feature specificity. Mean placement success is 6.2% for the position-only base, 13.8% with the previous action appended, 21.8% with explicit tracking error, and 26.6% with a compensated residual (Fig. 1). Tracking error exceeds the base by 15.6 points (Welch 95% CI 3.5 to 27.7). A position-history control that observes $[q_t, q_{t-8}]$ and no commands at all reaches 20.3% over three seeds, 14.1 points above the base (CI 4.5 to 23.8) and 1.5 points below tracking error (CI -11.1 to 14.0). Position history recovers most of the mean gain, but the interval establishes neither a difference nor equivalence. No pairwise contrast survives a Holm adjustment across all 28 pairs (Appendix F), so whether explicit tracking error is preferable to a suitable history is not resolved. Predicting tracking error as an auxiliary training target gives 6.6%: the channel helps as an input, not as a target.

A configured limit clips load telemetry. The SO-101's servos expose a load register excluded from our policy inputs. In the 50 demonstrations that register pins at its configured limit of 500 in 16.5% of frames, while tracking error keeps varying across those same frames (Appendix D); over all frames the two channels correlate at $r = -0.92$, and on a static bench with hanging masses the register is a linear function of tracking error ($R^2 = 0.976$). The stack's follower configuration caps gripper torque at half of maximum, register value 500, and the plateau sits at that cap. Tracking error retains variation at the telemetry plateau; this alone does not establish its physical-force accuracy.

Corpus: an observation, not an outcome label. A preregistered prediction held that larger tracking error at detected grasp onset would accompany successful grasps in archived teleoperation data. Sixteen public datasets with 546 episodes pass the inclusion gates. Only two meet the predicted effect-size threshold ($d > 0.5$); the pooled within-dataset effect is negative ($d = -0.39$) against a gripper-retention proxy, not independently labeled task outcomes. These records reject the proposed direction of association without identifying its cause (Appendix G).

5 Discussion and conclusion

The hardware gain and unresolved history contrast motivate three reporting items for the substrate beneath the policy. State the proprioceptive observation, including its history depth, since the default can conceal a consequential choice. Log applied commands, after any limiter, alongside measured positions, and distinguish the command rate from the policy's observation-update rate. Report actuator torque and current limits, since they can determine where telemetry saturates. Appendix I provides a filled reporting table and the study's analysis-plan provenance.

Three limits bound the claim. Three hardware seeds were trained; the two completed evaluations gave a 20-point and a 65-point improvement, the third was interrupted, and two completed seeds cannot bound that spread. The matched position-history control is registered, and its first hardware session was interrupted by a gripper servo fault (Appendix H), so the hardware result establishes the value of the channel, not its superiority over history. The hardware task never requires staying under

a force threshold, so nothing here shows learned force regulation. More seeds, the hardware history control, and a force-limited task are the next experiments.

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A Hardware protocol and complete evaluation record

Table 2: Full hardware evaluation record: success counts with Wilson 95% intervals in percentage points, discordant and concordant pair counts, exact McNemar p per seed, and the stratified exact test over the completed seeds. The interrupted seed has six recorded pairs, with 0/6 placements for each policy; its planned budget was 20 pairs. No confirmatory test is assigned to the incomplete comparison.

Seed	Evaluation	Base [95% CI]	Tracking error [95% CI]	tracking-error only / base only	both / neither	p
0	Complete	5/20 [+11, +47]	9/20 [+26, +66]	6 / 2	3 / 9	0.2891
1	Complete	1/20 [+1, +24]	14/20 [+48, +85]	14 / 1	0 / 5	0.0010
2	Interrupted	0/6 [+0, +39]	0/6 [+0, +39]	0 / 0	0 / 6	—
0+1	Stratified			20 / 3		0.0005

Table 3: Power of the exact two-sided McNemar test with 20 pairs at $\alpha = 0.05$, by the probability that a pair is discordant and the probability q that a discordant pair favours tracking error. Seed 0's observed structure (8 discordant pairs, 6 favouring) gives 0.17; seed 1's (15, 14) gives 0.97.

P(discordant)	$q = 0.7$	$q = 0.8$	$q = 0.9$	$q = 1.0$
0.2	0.02	0.05	0.10	0.20
0.4	0.11	0.26	0.53	0.87
0.6	0.19	0.46	0.80	1.00
0.8	0.29	0.64	0.93	1.00

Training. The recording contains 21,476 frames from 50 demonstrations at 30 Hz. Position and action statistics are computed on the recording; the training index then removes stationary prefixes, retaining 18,549 frames. Both policies use the same index and normalization, and recorded load channels are excluded from policy inputs. Gripper position is on the interface's 0–100 opening scale, not a calibrated force.

The ACT configuration uses a ResNet-18 image backbone, model width 256, four attention heads, feedforward width 1024, two encoder layers and one decoder layer. The action chunk and execution horizon are both 30 steps, with no temporal ensembling. Optimization uses AdamW at learning rate 10^{-4}, weight decay 10^{-4}, batch size 8 and gradient clipping at 10. All reported checkpoints are the final checkpoint at 6,000 steps. The observation projection necessarily changes width when six features are added; the remaining architecture and training settings are shared.

Execution and command semantics. Each rollout homes the arm to the mean demonstration start, replays the first 115 commands from demonstration 0, and hands control to the policy, so the reported task excludes autonomous approach from an arbitrary initial state. Each requested command passes through a relative-target limit of 8 degrees (8 opening units on the gripper)—a target-displacement bound rather than a force or velocity guarantee—and during both replay and policy execution the returned limited command supplies the next tracking-error calculation. The servo's native goal quantization sits below this software interface. State and images are read at each control step, but ACT consults them only when its 30-command queue is empty. Table 10 reports achieved timing from saved host timestamps; the nominal 30 Hz command loop does not imply 30 Hz policy feedback.

An offline replay check reproduces the training and inference state vectors on all 21,476 recorded frames for the base, tracking error, action history, position history and compensated residual designs. In seed 0, the saved tracking-error observations after the first rollout step also reproduce from logged applied commands, which checks software consistency, not physical command-to-encoder latency. Collection-time command limiting has not been established from the available metadata, so archived action labels should be read as recorded goals, which need not equal the goals applied to a servo if collection used an additional limiter.

Pairing and inference. Hardware resets are manual, and within each seed consecutive trials form a pair with AB/BA order alternating between pairs. The operator knows the policy identity, so outcomes are not blinded. Descriptively, seed-1 tracking error placed 7/10 whether it ran first or second

in a pair and in both halves of the session, and seed 0 placed 5/10 and 4/10 by order and 3/10 and 6/10 by half. Seeds 1 and 2 were scheduled after observing seed 0, with the seed-0 training recipe and evaluation budget fixed before their evaluations, and no policy was selected by hardware performance: every trained checkpoint was evaluated. The primary paired test is exact, two-sided McNemar with an improvement criterion of positive effect and $p < 0.05$; for the completed seeds, paired percentile bootstrap intervals (20,000 pair resamples, random seed 0) are $[-5, +45]$ and $[+40, +90]$ points, describing trial uncertainty for fixed checkpoints, not confidence intervals over retraining. The replication plan requested the equal-weight mean over all three seeds. Including the interrupted seed's observed zero difference gives 28.3 points; the 42.5-point mean in the main text describes the two completed seeds. Neither summary resolves training-seed variability, and the interrupted seed has a smaller trial budget and uncertain hardware integrity.

Interruptions and missing data. Seed 0 has 40 outcomes and 40 trajectories. Seed 1 has 40 outcomes and 39 trajectories: two rollouts, one per policy, completed their control horizon but reported a gripper overload while torque was being disabled; both are operator-labeled failures, and one trajectory file was lost to a logging bug that was fixed before evaluation resumed. Seed 2 was stopped after twelve labeled rollouts, all failures, forming six pairs; three reported the same overload at shutdown, and a thirteenth rollout completed its horizon but its outcome was not entered. We leave that outcome missing rather than infer it from joint motion, and we have no diagnosis of the recurring overload. In the seed-2 rollouts the jaw never closed in five of six position-only and three of six tracking-error trials, against three of 20 and one of 20 in seed 0 and two of 20 and none of 19 in seed 1 (Table 5), which is consistent with a gripper fault rather than a policy difference, although the traces cannot prove it. Trial-level detail is in the supplementary evaluation records.

Gripper overloads by policy. Table 4 counts, for every hardware evaluation record, the rollouts of each policy that ended with the servo library's gripper-overload error while torque was being disabled (process abort), those that failed for another reason, and those whose outcome was never entered. Every such rollout had completed its control horizon. The seed-0 record predates the shutdown-status field; its runner wrote the trajectory only after torque release, so a fault would have left a missing trajectory, and all 40 are present. The 2026-09-04 residual comparison predates the field as well and is not tabulated. Across all records the fault occurred in 4 of 47 position-only, 6 of 55 tracking-error and 0 of 10 position-history rollouts. Within the interrupted history session it struck 4 of 9 tracking-error rollouts and 0 of 10 position-history rollouts (post hoc Fisher exact $p = 0.03$); the tracking-error policy is the one that closes and holds the jaw to the end of the horizon (Table 5), so the pattern is consistent with the fault being raised when torque is released from a stalled, loaded jaw, but the servo's condition also changed over that session, and no diagnosis has been made.

Table 4: Gripper overload incidents per policy. Rollouts counts labeled outcomes plus any completed rollout whose outcome was not entered. Overload: the servo library raised its overload error while torque was being disabled after a completed horizon. Other failed exit: a non-overload process failure (one logging fault). Generated from the trial records and incident files under `results/hardware/`.

Evaluation record	Policy	Rollouts	Overload at shutdown	Other failed exit	Outcome missing
Tracking error vs. position, seed 0	Position only	20	0	0	0
	Tracking error	20	0	0	0
Tracking error vs. position, seed 1	Position only	20	1	0	0
	Tracking error	20	1	0	0
Tracking error vs. position, seed 2 (interrupted)	Position only	7	3	0	1
	Tracking error	6	1	0	0
Tracking error vs. position history, seed 1 (interrupted)	Tracking error	9	4	0	0
	Position history	10	0	1	0
All records	Position only	47	4	0	1
All records	Tracking error	55	6	0	0
All records	Position history	10	0	1	0

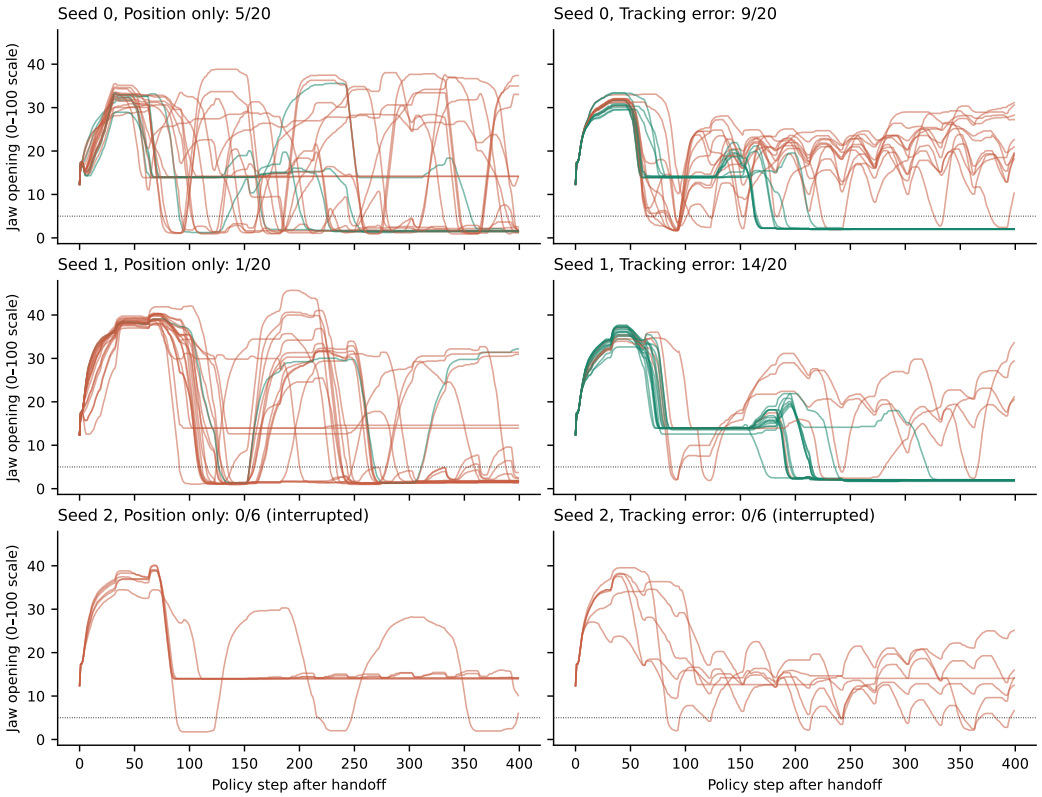


Figure 3: Recorded jaw position for every hardware rollout, by seed and policy; green traces placed the adapter, orange did not. Policy control begins at step 0 after the scripted replay. The dotted line marks the closure threshold used in Table 5.

Table 5: Jaw-trace classes for all recorded rollouts. Never closed: jaw never below 5 recorded units. Reopened: an excursion above 8 units after closing. Units are the gripper's 0–100 opening scale. Closure step is the median first step below the threshold. One seed-1 tracking-error trajectory is missing.

Seed	Policy	Outcome	n	never closed	closed, reopened	closed, held	median step of closure
0	Position only	success	5	0	1	4	203
0	Position only	failure	15	3	8	4	110
0	Tracking error	success	9	0	0	9	165
0	Tracking error	failure	11	1	10	0	83
1	Position only	success	1	0	1	0	127
1	Position only	failure	19	2	13	4	116
1	Tracking error	success	14	0	0	14	206
1	Tracking error	failure	5	0	4	1	115
2	Position only	failure	6	5	1	0	88
2	Tracking error	failure	6	3	3	0	241

B Exploratory trajectory analysis

Every recorded rollout was classified from its jaw trace as never closed, closed then reopened, or closed and held (Table 5, Fig. 3). All 23 tracking-error successes in seeds 0 and 1 closed once and held to the end of the horizon, and 15 of the 16 failures with a recorded trajectory either never closed or reopened (one seed-1 failure has no trace). Position-only successes are too few to characterize. Successful tracking-error rollouts closed later, at median step 165 and 206 against 83 and 115 for failures, and in seed 0 they requested smaller moves: the median fraction of steps altered by the command limiter is 0.09 for tracking error and 0.32 for the base, while in seed 1 the fractions

are 0.32 and 0.35. Over the last 100 steps of the seed-0 tracking-error rollouts, median summed joint-coordinate travel, a descriptive sum over mixed coordinates, is 14.3 for successes and 93.6 for failures. Later closure and less motion in successful trials could simply follow from progressing further through the task. Traces were grouped by outcome after the fact and contain no video, object pose or force, so they cannot say whether a failure was a missed grasp, a slip or a misplacement, and none of these regularities is a validated success detector. In tracking-error rollouts the median absolute jaw tracking error after closure is 0.11 and 0.16 units for successes and 1.04 and 0.88 for failures (seeds 0 and 1): a held grasp settles to a small steady discrepancy, whereas reopening and re-closing keeps it large. Fig. 4 shows when the jaw first closes in each rollout and, for the demonstrations, when the grasp closure occurs relative to the replayed prefix.

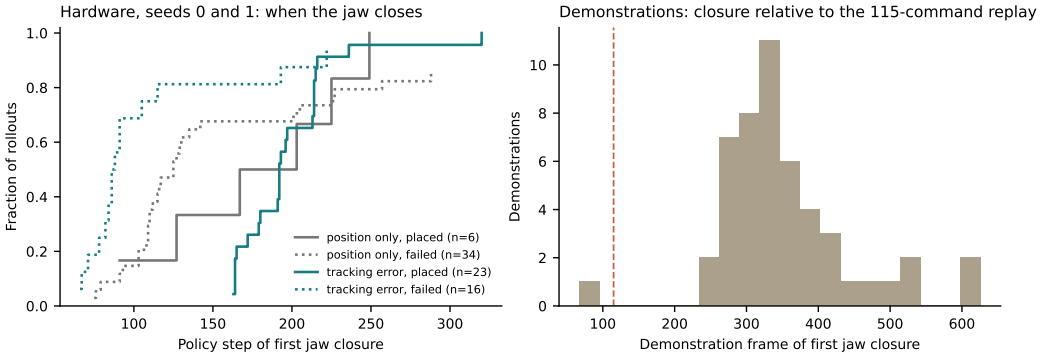


Figure 4: Left: cumulative fraction of hardware rollouts (seeds 0 and 1) whose jaw has closed by a given policy step, by policy and outcome. Right: frame of the grasp closure in the 49 demonstrations that close; the dashed line is the 115-command replay.

C Checkpoint sensitivity and offline error

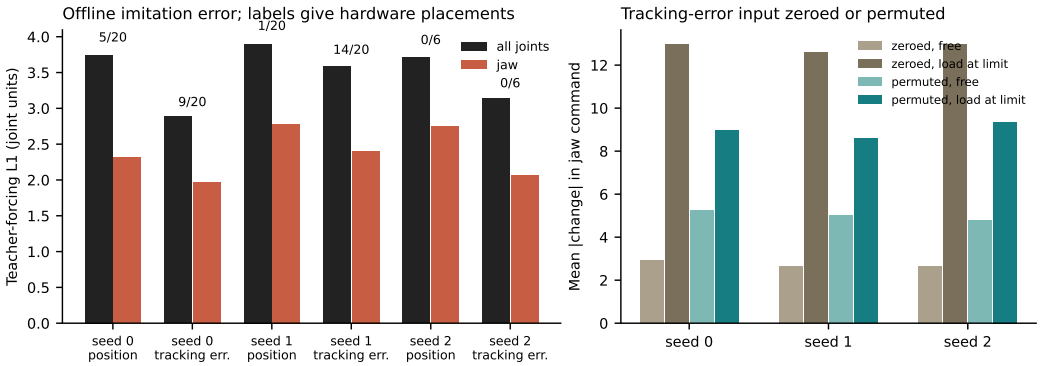


Figure 5: Left: teacher-forcing L1 error of each evaluated checkpoint's predicted 30-command chunk on every fourth training frame (4,638 frames), all joints and jaw; labels give hardware placements. Right: mean absolute change in the predicted jaw command when the six tracking-error inputs are zeroed or permuted across frames, split by whether the recorded gripper load register is at its limit (881 frames) or not.

The hardware trials cannot say whether a policy reads its tracking-error input, so the six evaluated checkpoints were re-scored offline on the demonstration frames (Fig. 5). Zeroing the input changes the predicted jaw command by 12.6 to 13.0 units where the load register is pinned and by 2.7 to 2.9 units on other frames; permuting it across frames changes the command by 8.6 to 9.4 and 4.8 to 5.2 units. The mean recorded command step is 0.5 units, and teacher-forcing error rises from 2.9–3.6 to 6.2–6.8 under zeroing. All three tracking-error checkpoints therefore depend on the channel,

most on load-register-saturated frames; this is a property of the networks, not evidence that the dependence is what produced the hardware placements. Offline error itself is a fit diagnostic on training frames, since no demonstrations were held out: it separates designs but ranks the seed-0 tracking-error checkpoint (2.9, 9/20) below the seed-1 checkpoint (3.6, 14/20). Training losses of the seed-1 and seed-2 checkpoints track each other closely (Fig. 6).

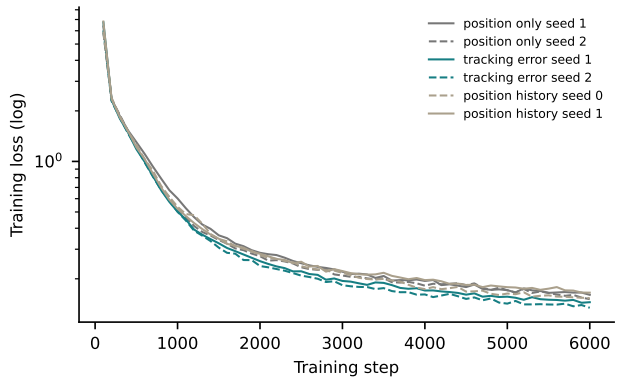


Figure 6: Training loss, logged every 100 steps, for the checkpoints whose logs are in the supplementary material.

D The shipped load register saturates at a configured torque limit

A separate static experiment on one shoulder-lift STS3215 uses twelve hanging masses at five poses (60 measurements). Across those measurements, the fitted relationship between the load register and tracking error is $\texttt{Present_Load} = -4.70\delta + 0.1$, with $R^2 = 0.976$. Its relation to applied mass is far less stable, with per-pose R^2 for tracking error ranging from 0.085 to 0.910, so the tight load-register association does not establish an equally accurate physical-load measurement.

In the 50 teleoperation demonstrations, the gripper load register reaches 500 in every episode and remains there in 16.5% of frames, while tracking error continues to vary across those same frames (Fig. 7); over all frames the two channels correlate at $r = -0.92$. The inspected follower configuration sets a reduced gripper torque limit, consistent with that observed plateau. This motivates inspecting the command–position discrepancy when telemetry saturates, but it does not calibrate the remaining discrepancy in newtons: no jaw load-cell calibration is available, and the simulator's force conversions describe the simulated plant only.

E Simulated plant

The simulated plant was characterised with scripted experts before any policy was trained, to check that tracking error can carry contact information in this simulator at all. Three experts differ only in the signal that sets the grip: a fixed jaw angle (clamp), the quantised tracking error, or the true contact force (oracle). Without a crush limit the clamp places 29 of 30 and the two regulated experts 22 and 20, because a gentler grip drops more often and nothing penalises crushing. With a crush limit the clamp fails every episode at a median peak grip of 355 N, while the tracking-error expert at the native quantum places within a few episodes of the oracle at every limit, and coarsening the observation quantum destroys the task as a cliff whose position moves with the margin (Table 6). This is why the learning grid could ask whether a policy uses the channel: in this plant the channel suffices for force regulation by a scripted controller at the real encoder's resolution. Simulation tracking error is the actuator target minus the encoder reading, both rounded to integer counts; the position actuators use proportional gain 50 with force limited to ± 35. It does not show that the physical servo provides the same information; Appendix D addresses that separately.

Table 6: Scripted-expert placements out of 30 randomised episodes in the simulated plant, by grip signal and observation quantum, with and without a crush limit that fails the episode if grip force exceeds it.

	tracking error, quantum q (counts)					clamp	oracle
Crush limit	1	4	8	16	32		
no limit	22	—	—	—	—	29	20
100 N	11	11	9	1	0	0	12
120 N	17	17	16	10	0	0	14
160 N	20	20	19	17	11	0	17

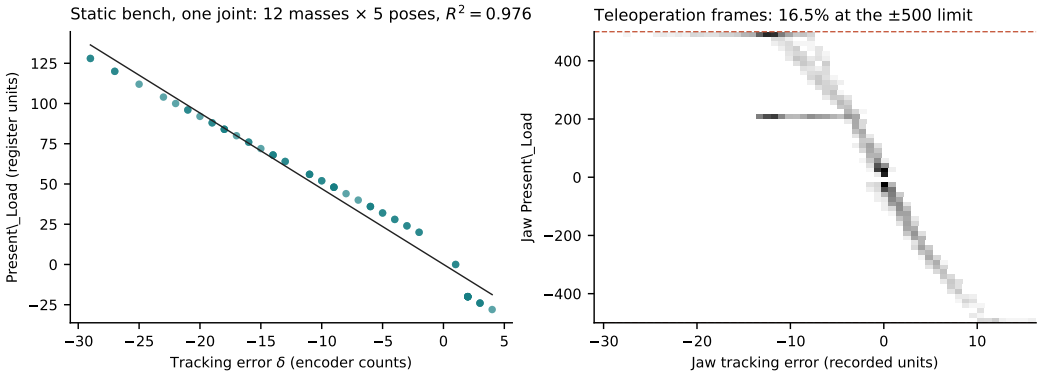


Figure 7: Left: static bench, one shoulder-lift servo, load register against tracking error over twelve hanging masses at five poses, with the fitted line. Right: jaw load register against jaw tracking error on the teleoperation frames (log-count shading); the dashed lines are the configured ± 500 torque limit.

F Additional simulation and hardware comparisons

Simulation recordings use integer encoder counts at 15 Hz, obtained by subsampling the 30 Hz controller. The scripted teacher regulates the jaw using tracking error, and the successful demonstration set includes randomized grip depth and recovery perturbations. These collection choices are shared across observation designs, of which the position-history control observes $[q_t, q_{t-8}]$ and one-command history observes $[q_t, g_{t-1}]$. Neither comparison makes the simulation and hardware tasks identical: camera views, demonstrations and recording cadence all differ between the two settings.

The original simulation study includes an auxiliary tracking-error target and a latched seat token in addition to the designs in Fig. 1, with mean placement rates of 6.6% and 21.8% across five seeds; the seat token has the same mean as tracking error. The compensated residual is $e_t = \delta_t - \widehat{K}(g_{t-1} - g_{t-2})$, where each joint's coefficient is fitted once on presumed free-motion demonstration frames. A two-command history control reaches 14.7% over three seeds; comparing it with the residual holds available raw information fixed but does not match input width. Under the original study's 15-point practical threshold at three training seeds, neither this comparison nor the residual–position-history comparison resolves at three seeds, and the stronger feature-specific interpretation remains open. Table 7 lists the main design contrasts with Welch intervals; the Holm column adjusts across all 28 pairwise comparisons, and no contrast survives that adjustment. The original plan predicted the new designs would improve on action history, and was written after the first three tracking-error results existed. The seed-count extrapolation $15\sqrt{3/n}$ displayed in the table and the all-pairs statistical analysis are retrospective; the threshold is a descriptive effect-size rule, not a significance test (Appendix I).

An earlier hardware comparison, run before the tracking-error study, evaluated the compensated residual: the base placed 7/10 objects and the residual 0/10 (registered exact test on consecutive trials from the same reset: 7 discordant pairs, all favouring the base, $p = 0.016$; Fisher exact

Table 7: Simulation design contrasts in placement-success points, with Welch 95% intervals, unadjusted and Holm-adjusted p values across all 28 pairs, the retrospective resolution heuristic $15\sqrt{3/n}$ for the smaller seed count, and whether the difference clears it.

Contrast	Points	Welch 95% CI	p	Holm p	Rule	Clears
Position history − Position	+14.1	[+4.5, +23.8]	0.013	0.33	15.0	no
Action history − Position	+7.6	[-2.5, +17.7]	0.121	1.00	11.6	no
Tracking error − Position	+15.6	[+3.5, +27.7]	0.019	0.44	11.6	yes
Compensated residual − Position	+20.4	[+10.2, +30.6]	0.002	0.05	11.6	yes
Auxiliary target − Position	+0.4	[-7.1, +7.9]	0.903	1.00	11.6	no
Tracking error − Position history	+1.5	[-11.1, +14.0]	0.784	1.00	15.0	no
Compensated residual − Position history	+6.3	[-4.6, +17.1]	0.206	1.00	15.0	no
Tracking error − Action history	+8.0	[-4.9, +20.9]	0.188	1.00	11.6	no
Compensated residual − Two-command history	+11.9	[+2.2, +21.7]	0.025	0.49	15.0	no
Compensated residual − Tracking error	+4.8	[-8.1, +17.7]	0.413	1.00	11.6	no

$p = 0.003$; Wilson intervals 40 to 89% and 0 to 28%; same 115-command replay). The coefficients fitted from these real demonstrations differ markedly from the simulation fits, and a free-motion labeling heuristic includes contact frames. Both are plausible sources of mismatch, not a diagnosis. Coefficients above one do not by themselves invalidate the regression, and that runner also fed back requested rather than post-limit commands, so preprocessing and execution both differ from the tracking-error study.

The simulation task admits a crush threshold, but the learning grid does not activate it. A post-hoc screen of existing checkpoints at 120 and 60 N produces heavy crushing and near-floor success, failing the predicted advantage of the compensated residual. A separate runtime guard reduces crush events from 85 to 3 across 180 crush-tier episodes, while eliminating all 12 unguarded placements across its broader 270-episode comparison. Neither result is learned force regulation, and the guard was not used on hardware.

Larger-budget experiments do not establish a scaling advantage. An earlier 600-demonstration comparison of tracking error and one-command history reaches 57% and 60%, respectively, over two seeds at 224 pixels and 50,000 steps. A later 600-demonstration screen saved at 26,000 steps contains three base and three residual evaluations, each with 0/100 placements. These floor results are reported separately from the 280-demonstration grid; they support neither a positive scaling claim nor equivalence between observations.

G Archived teleoperation study

The preregistered corpus prediction was that tracking error at a detected seat event would be larger in proxy-successful episodes, with Cohen's $d > 0.5$ in at least ten of the twenty datasets targeted for scanning; 77 were scanned. Sixteen datasets and 546 episodes pass the inclusion gates, including 283 proxy successes under a success proxy of continued gripper closure through transport. Only two datasets exceed the positive threshold, and the pooled within-dataset effect is $d = -0.39$—a failure of the proposed direction and labeling rule, though not proof that no function of command and position history can predict outcome. The measured feature is the mean absolute jaw discrepancy over eight frames after a detected stall. Inclusion requires at least 20 eligible episodes, five per proxy class, and a median of no more than one closure. The proxy requires no early reopening and at least 60 encoder-equivalent counts of arm displacement. Both feature and proxy derive from the same kinematic traces; no independent object-outcome annotation validates the proxy. The registered secondary directional AUC is 0.39, also below its predicted 0.5 threshold. Eleven of the sixteen point estimates are negative and seven of those intervals exclude zero; two are positive with intervals excluding zero (Fig. 8).

Empty closure against a mechanical stop and closure on an object can both produce a seat event, and a large command discrepancy can equally reflect an operator continuing to close against resistance. These ambiguities explain why a contact-related feature need not be a success label, although the

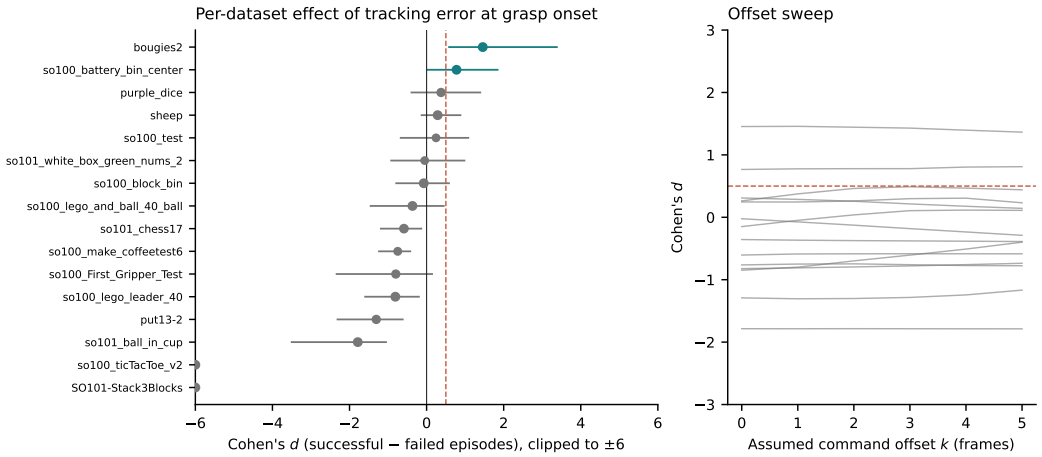


Figure 8: Left: per-dataset Cohen's d of tracking error at grasp onset between proxy-successful and failed episodes, with 95% intervals, clipped to ± 6; marker area scales with episode count and the dashed line is the preregistered $d = 0.5$ threshold. Right: the same effect sizes under assumed command offsets of zero to five frames.

corpus does not isolate their causal contributions. Recorded leader commands may differ from applied follower goals and the timing is only approximately aligned, but an offset sweep from zero to five recorded frames leaves the qualitative conclusion unchanged, with median per-dataset range of d across offsets 0.12.

H Position-history control on hardware

The simulation grid leaves open whether the hardware gain from tracking error is specific to the command or would follow from any temporal input. A matched position-history control, $[q_t, q_{t-8}]$ with the training recipe of Appendix A, is pre-registered in the supplementary material together with the two paired comparisons that would test it: tracking error against position history for seed 1, and position only against position history for seed 0, each with 20 pairs under the protocol of Appendix A. The initial seed-1 evaluation was interrupted after 9 complete pairs (19 rollouts, 4 ending with a gripper overload error): the gripper servo stopped following jaw commands in several rollouts and the overhead camera was found displaced relative to the earlier sessions. Its outcomes are retained (Table 8), and no confirmatory test is assigned to it. A separate logging failure after normal torque release accounts for the fifth failed exit; its operator-labeled outcome is retained despite the missing trajectory. The seed-0 history checkpoint is trained, not evaluated. The registered hardware history control remains incomplete.

Table 8: Pre-registered position-history comparisons on hardware; A is the first-named policy. Each evaluation record is shown separately. Errors count failed process exits, including a logging failure; incomplete comparisons have no test.

Seed (record)	Comparison (A vs. B)	Status	A	B	B-only / A-only	p	errors
1 (v1)	Tracking error vs. position history	Interrupted	1/9	0/9	0 / 1	—	5

I Design-choice reporting and analysis-plan provenance

Table 9 records the choices needed to interpret this comparison. It separates configured values, measurements, and unresolved metadata; it is a worked reporting artifact for this study, not a validated universal checklist.

Table 9: Design choices beneath the policy in the hardware comparison.

Choice	Setting and evidential boundary
Observation	One overhead RGB image resized to 96×96; six joint positions; six additional tracking-error features for the intervention. Load registers are recorded but excluded from the policy.
Coordinates	Arm positions in degrees; jaw on a 0–100 opening scale. Positions use dataset standardization; discrepancy is divided by 8.
Command semantics	Absolute position goals. Deployment logs the requested and returned post-limit goals; collection-time limiting is unverified.
Limiter and telemetry	Relative-target limit 8 in the above coordinates; inspected gripper torque cap 500. The limit bounds target displacement; neither it nor telemetry is a calibrated jaw-force measurement.
Time and history	Demonstrations nominally 30 Hz; 30 commands executed per policy query, without temporal ensembling. The history control uses eight recorded frames (0.27 s nominally; 0.53 s in simulation).
Demonstrations	50 episodes, 21,476 frames; 18,549 training frames after stationary-prefix trimming. No held-out demonstrations in the offline checkpoint diagnostic.
Reset and scope	Manual nominal object reset; replay 115 commands from demonstration 0, then up to 400 policy steps. One adapter-and-roll task.
Evaluation unit	AB/BA pairs within each training seed, operator-labeled and unblinded. Two complete 20-pair comparisons; six interrupted pairs for the third seed. Trial uncertainty is conditional on checkpoints.
Missing evidence	No jaw load cell, physical latency calibration, or ground-truth object traces. Hardware history control remains incomplete.

Table 10: Achieved hardware timing from all 91 saved labeled-trial trajectories in the original tracking-error comparison. Rates are medians of per-rollout rates $(N-1)/(t_{N-1}-t_0)$. Intervals are pooled medians between host timestamps recorded after command submission, split at 30-command chunk boundaries. These are not bus latency or encoder sample-age measurements. One seed-1 trajectory is missing.

Seed	Input	Traces	Rate (Hz)	Within chunk (ms)	At boundary (ms)
0	Position	20	29.47	33.3	73.2
0	Tracking error	20	29.49	33.4	72.6
1	Position	20	29.49	33.3	73.0
1	Tracking error	19	29.48	33.3	73.2
2	Position	6	29.48	33.3	73.8
2	Tracking error	6	29.57	33.4	69.2

The original simulation design note and corpus operational definitions are included as archived files in the supplementary artifact. The simulation note was recorded on 17 August after three tracking-error evaluations, before the auxiliary-target, residual, and token implementations. It specified a 15-point threshold at three seeds and predicted those designs would improve on action history. It did not preregister the later position-history comparison, the seed-count extrapolation, or the all-pairs Welch/Holm analysis. The corpus definitions were recorded on 16 August before the run; the archived source and implementation specify the retention proxy and selection gates. The seed-0 hardware plan is reported by its author as written before training, but has no independently timestamped artifact preceding that session. The replication plan has a source hash in the pre-training manifest. The hardware history control was registered after the simulation history results. The stratified hardware test, checkpoint interventions, and trajectory/timing summaries are post hoc.